

Offline Reinforcement Learning for Sepsis Treatment COMP 579 Final Project Report (Draft)

Alex Groiser

Yusuf Hamza

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1 Motivation and Problem Setting

Sepsis remains a major cause of mortality in intensive care units. Clinical treatment requires sequential decisions over intravenous fluids and vasopressors under substantial uncertainty. This is naturally modeled as a finite-horizon Markov Decision Process (MDP), where states represent patient physiology, actions represent treatment decisions, and terminal outcomes reflect survival-oriented objectives. Because online experimentation is unsafe in this domain, we target *offline reinforcement learning* (offline RL): learning solely from historical logged trajectories.

Our project goal is to evaluate whether a conservative offline RL policy can outperform a behavior-cloning baseline under off-policy evaluation (OPE), while maintaining methodological discipline around data leakage, support mismatch, and reproducibility.

2 Related Work and Course Connection

Our approach is grounded in the offline RL literature and directly matches COMP 579 content on off-policy learning and policy evaluation.

- **Conservative Q-Learning (CQL)** penalizes unsupported out-of-distribution actions and is designed for stable offline value learning (Kumar et al., 2020).
- **Behavior Cloning (BC)** provides an imitation baseline and reflects the logged clinician policy.
- **Weighted Importance Sampling (WIS)** provides an off-policy value estimator with lower variance than ordinary IS in many practical settings.
- The pipeline aligns with lecture themes on off-policy RL, policy evaluation, and offline/batch RL (Feb 12, Mar 19, Mar 26 in the course schedule).

3 Data and Cohort Construction

3.1 Data Source

We implemented and validated the full pipeline on the publicly available `mimic-iv-clinical-database-demo-on-fhir-2.0` FHIR subset. This serves as an engineering and methodological proxy before running on full credentialed MIMIC-IV data.

3.2 Sepsis Cohort Logic

We select ICU encounters (`EncounterICU`) whose linked conditions include sepsis-related ICD patterns (e.g., A40/A41, 038, textual sepsis terms). Because FHIR encounter references may not always align one-to-one with ICU stay IDs, events are mapped to ICU windows by patient and timestamp when needed.

3.3 State, Action, Reward

- **State:** top 16 frequent numeric lab features from `ObservationLabevents` (code-frequency based), represented as hourly-updated vectors.
- **Action:** discretized 2D treatment index from fluid amount and vasopressor intensity bins.
- **Reward:** terminal survival proxy (+1 if alive at 30-day window, -1 if deceased), with 0 intermediate rewards.

3.4 Methodological Safeguards

- Patient-level **train/val/test** split (no transition-level mixing).
- Behavior policy probabilities estimated from **train split only** (Laplace-smoothed frequency estimator) to reduce leakage.
- Reproducible seeds and script-based execution for all stages.

4 Methods

4.1 Offline RL Algorithms

BC baseline: discrete behavior cloning with MLP encoder via `d3r1py`.

CQL: discrete CQL with conservative penalty coefficient α and fixed optimizer/architecture defaults from `d3r1py`.

4.2 Evaluation Protocol

We use Weighted Importance Sampling (WIS) with bootstrap confidence intervals:

$$\hat{V}_{\text{WIS}} = \frac{\sum_i w_i G_i}{\sum_i w_i}, \quad w_i = \prod_t \frac{\pi(a_t | s_t)}{\mu(a_t | s_t)}$$

where μ is the estimated behavior policy and π is the candidate policy. We evaluate on held-out split data and report 95% bootstrap CIs.

5 Implementation and Reproducibility

Pipeline stages are script-based:

- Data extraction: `src/data/extract_sepsis_cohort.py`
- Dataset export: `src/data/build_mdp_dataset.py`
- Training: `src/train/train_bc.py`, `src/train/train_cql.py`
- OPE: `src/ope/wis_eval.py`
- End-to-end runner: `experiments/run_all.py`

The project includes compatibility handling for `d3r1py` API differences (dataset loading and constructor signatures), enabling consistent execution across local environments.

6 Results on Public Demo FHIR Data

Using the current pipeline outputs:

- Cohort ICU encounters: 30
- Sepsis patients: 16
- Transitions: 1320
- State dimension: 16

OPE summary (WIS, 95% CI):

Policy	Mean Return	95% CI Low	95% CI High
Behavior (logged, test split)	-0.500	–	–
BC (proxy policy probs)	-0.315	-1.000	0.587
CQL ($\alpha = 1.0$, proxy policy probs)	-0.424	-1.000	0.976

Split diagnostics: train/val/test transitions are 1020/126/174 respectively, with only 4 held-out test episodes in this demo subset.

Interpretation: On this small demo subset, uncertainty is substantial and confidence intervals overlap heavily. Results are therefore inconclusive and mainly validate pipeline correctness rather than clinical superiority claims.

7 Open-Benchmark Validation: ICU-Sepsis

To reduce schedule risk while waiting for credentialed data, we added an open benchmark track using **Sepsis/ICU-Sepsis-v2**. This provides a reproducible, no-approval-required environment for validating algorithm and evaluation plumbing.

Observed benchmark summary (current run):

- Episodes: 50
- Transitions: 359
- State dimension: 716 (discrete-state one-hot encoding)
- Test split episodes: 8

WIS summary on ICU-Sepsis test split:

Policy	Mean Return	95% CI Low	95% CI High
Behavior (logged)	0.750	–	–
BC (proxy policy probs)	0.772	0.429	1.000
CQL ($\alpha = 1.0$, proxy policy probs)	0.786	0.498	1.000

Interpretation: ICU-Sepsis provides a stronger algorithmic smoke test than tiny FHIR subsets and helps verify that BC/CQL/OPE components are functioning consistently in a controlled benchmark setting. However, benchmark gains do not by themselves establish clinical validity on real MIMIC-IV.

8 Limitations and Validity Threats

- The demo FHIR subset is not representative of full MIMIC-IV scale.
- OPE policy probabilities for BC/CQL are currently proxy-level approximations in this draft pipeline; final submission should replace this with stronger policy-probability estimation or additional doubly robust analyses.
- Action discretization and state selection are pragmatic first-pass choices and require sensitivity analysis.

9 Planned Final Experiments on Credentialed MIMIC-IV

1. Finalize sepsis cohort protocol and feature map with train-only normalization.
2. Run CQL hyperparameter sweeps (α and learning rate) across multiple random seeds.
3. Report subgroup and split sensitivity checks.
4. Extend OPE robustness (clipping sweeps, variance analysis, optional doubly robust estimator).

10 Contributions

- **Alex Groiser:** data engineering, FHIR cohort extraction, MDP construction, reproducibility scaffolding.
- **Yusuf Hamza:** training/evaluation integration, OPE pipeline, experiment orchestration and analysis.
- **Shared:** report writing, interpretation, final validation and presentation.

References

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